

loss f^n : regularized
differentiable.

XGBoost for Regression

cgpa	package
6.7	4.5
9.0	11.0
7.5	6.0
5.0	8.0

Working principle
same as Gradient
Boosting.

But, in Xg boost the decision tree is different.

cgpa	package	pred model 1	residual 1	pred model 2	residual 2
6.7	4.5	7.3	-2.8	6.69	-2.19
9.0	11.0	7.3	3.7	8.41	2.59
7.5	6.0	7.3	-1.3	6.69	-0.69
5.0	8.0	7.3	0.7	7.51	0.19

residual
is decreasing

mean

decision tree by cgpa + Residual 1.

$[-2.8, 3.7, -1.3, 0.7]$ → Similarity score (SS)

$$SS = \frac{(\sum \text{residuals})^2}{\# \text{residuals} + \lambda} \rightarrow \text{regularization.}$$

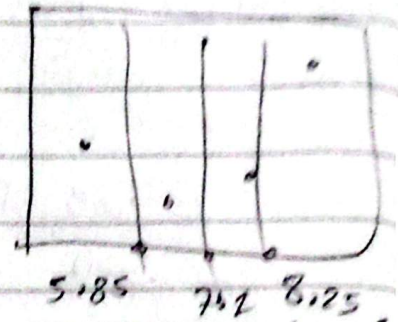
$$L=0$$

$$SS_{\text{root}} = \frac{(-2.8 + 3.7 - 1.3 + 0.7)^2}{4+0} = \frac{0.3^2}{4} = 0.02$$

Now, sort the feature in ascending.

$$\left. \begin{array}{l} 5.0 \\ 6.7 \\ 7.5 \\ 9.0 \end{array} \right\} \left\{ \begin{array}{l} 5.85 \\ 7.1 \\ 8.25 \end{array} \right\} \left\{ \begin{array}{l} \text{splitting} \\ \text{criteria.} \end{array} \right.$$

mean of 2



which gives better ~~split~~ improvement becoming root node.

Let $L=0$ for now.

Splitting criteria

① 5.85

$$c_{gpa} < 5.85$$

$$\left\{ \begin{array}{l} \text{leaf 1} \\ 0.7 \end{array} \right\} \quad \left\{ \begin{array}{l} \text{leaf 2} \\ -2.8, -1.3, 3.7 \end{array} \right\}$$

$$SS = \frac{(\sum \text{residuals})^2}{\# \text{residual} + L}$$

$$SS_L = \frac{0.7^2}{1+0} = 0.49$$

$$SS_R = \frac{(-2 - 1.3 + 3.7)^2}{3+0} = 0.05$$

$$\text{gain} = (SS_L + SS_R) - SS_{\text{root}} = 0.52$$

Splitting criteria

② 7.1

$$c_{gpa} < 7.1$$

$$\left\{ \begin{array}{l} \text{leaf 1} \\ 0.7, -2.8 \end{array} \right\} \quad \left\{ \begin{array}{l} \text{leaf 2} \\ -1.3, 3.7 \end{array} \right\}$$

$$SS_L = \frac{(0.7 - 2.8)^2}{2+0} = 2.20$$

$$SS_R = \frac{(-1.3 + 3.7)^2}{2} = 2.88$$

$$\text{gain} = (SS_L + SS_R) - SS_{\text{root}} = 5.06$$

Splitting criteria

③ 8.25

$$c_{gpa} < 8.25$$

$$\left\{ \begin{array}{l} \text{leaf 1} \\ 0.7, -2.8, -1.3 \end{array} \right\} \quad \left\{ \begin{array}{l} \text{leaf 2} \\ 3.7 \end{array} \right\}$$

$$SS_L = 3.85$$

$$SS_R = 13.65$$

$$\text{gain} = 17.52$$